



Robotic Grasping and Manipulation An Introduction

Miao Li

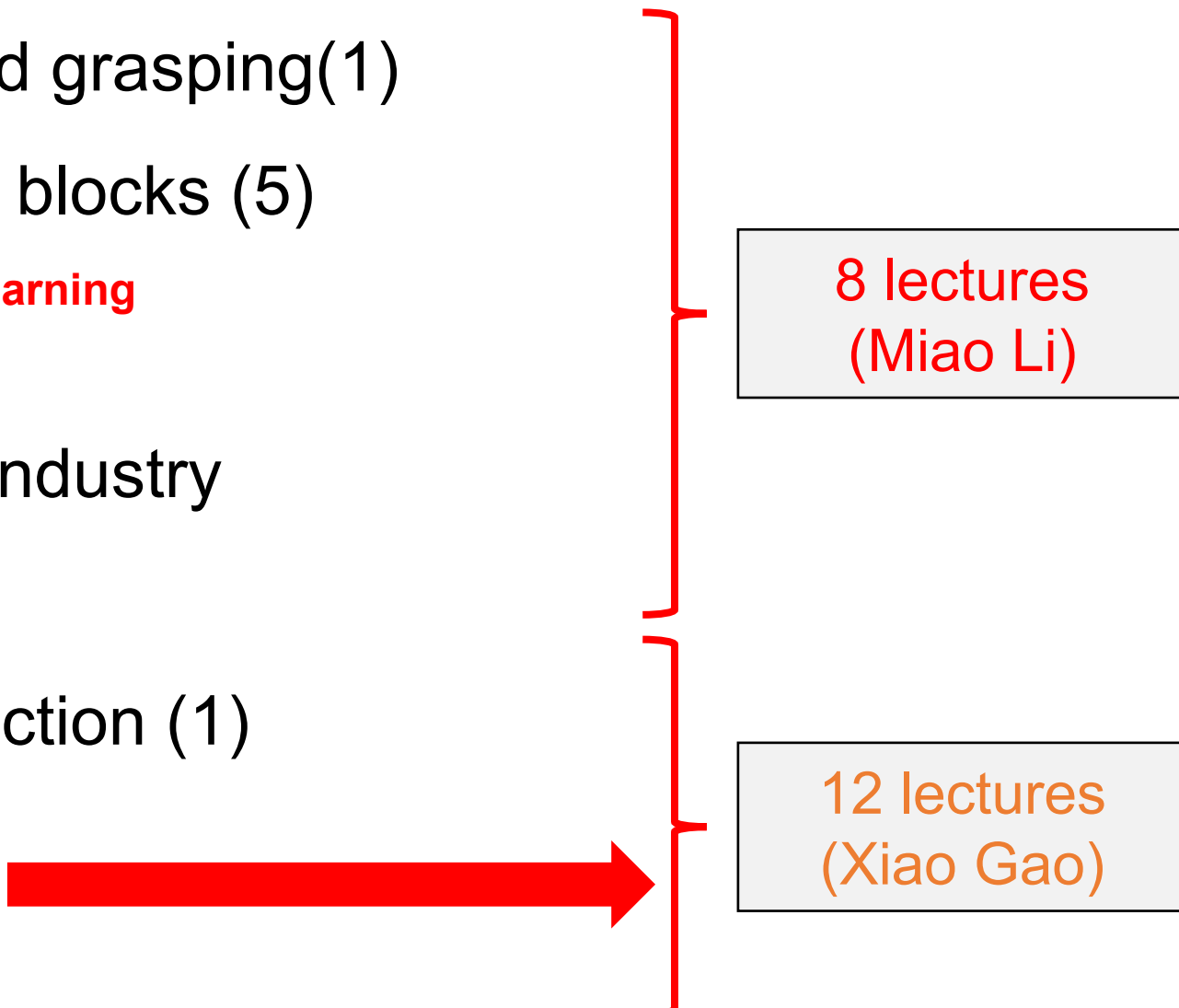
2025, Wuhan University

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Aims of this lecture

- Introduction to robotic hand and grasping(1)
- Introduction to the key building blocks (5)
 - **Design, Perception, Planning, Control, Learning**
 - Deep learning, optimization....
- Perspective from academia & industry
& paper reading (1)
- Paper reading and project selection (1)
- Hand design and simulation
- Hands-on practices and Q&A



8 lectures
(Miao Li)

12 lectures
(Xiao Gao)

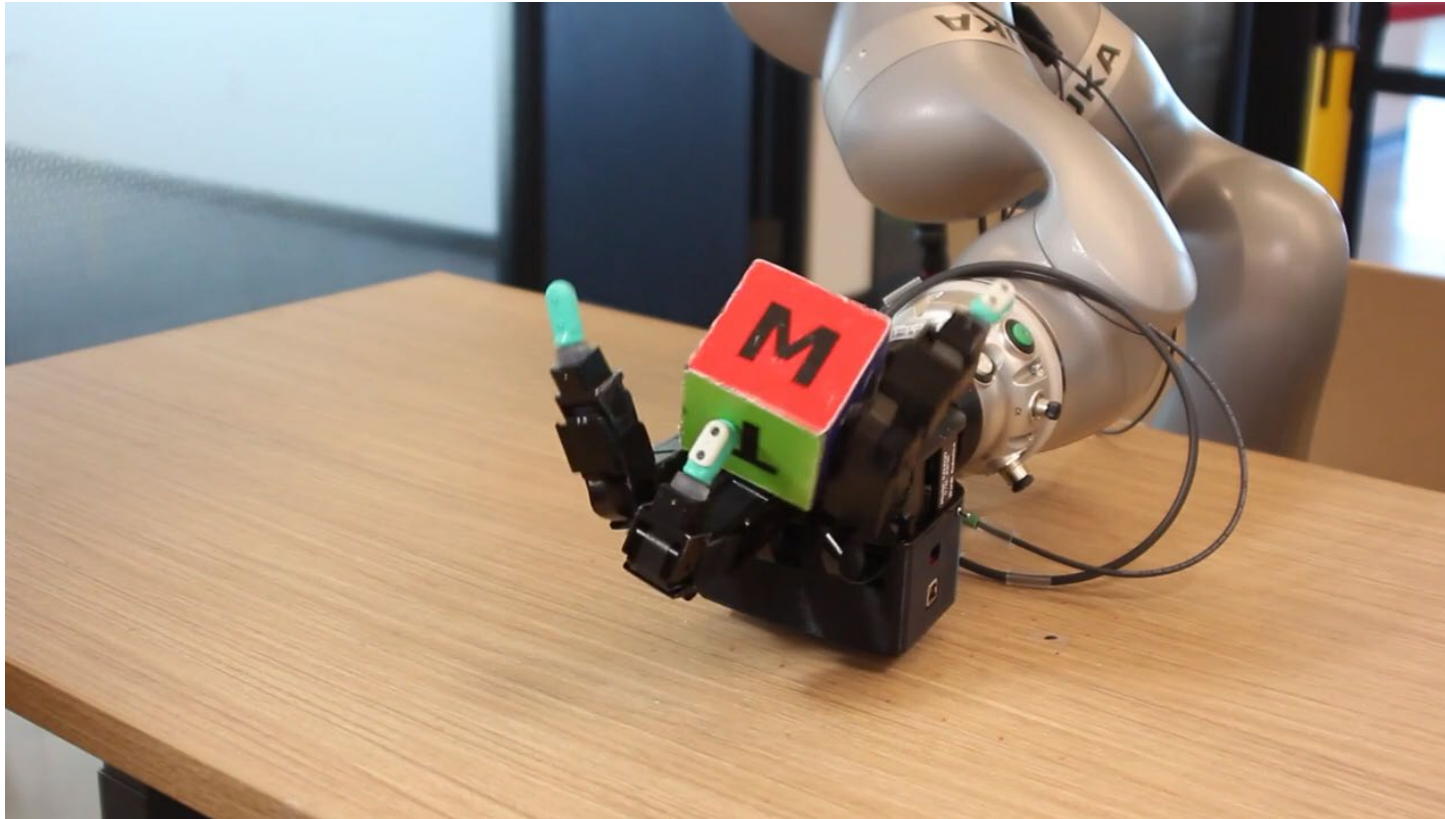
What are the core problems in this area?

- How to design a robotic hand for a specific application?
- How to plan a suitable/optimal grasp?
- How to evaluate a grasp or a manipulation?
- How to design a human-like tactile sensor?
- How to control a robotic hand to perform some fine manipulations?
- Data driven or Model-based?
- How to achieve human-like dexterous skills?
- Is there a unified model for dexterous manipulation?

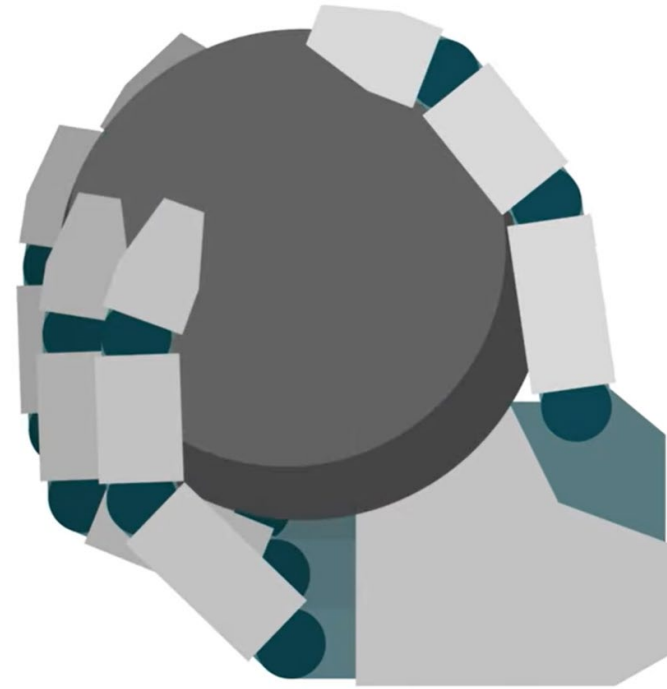
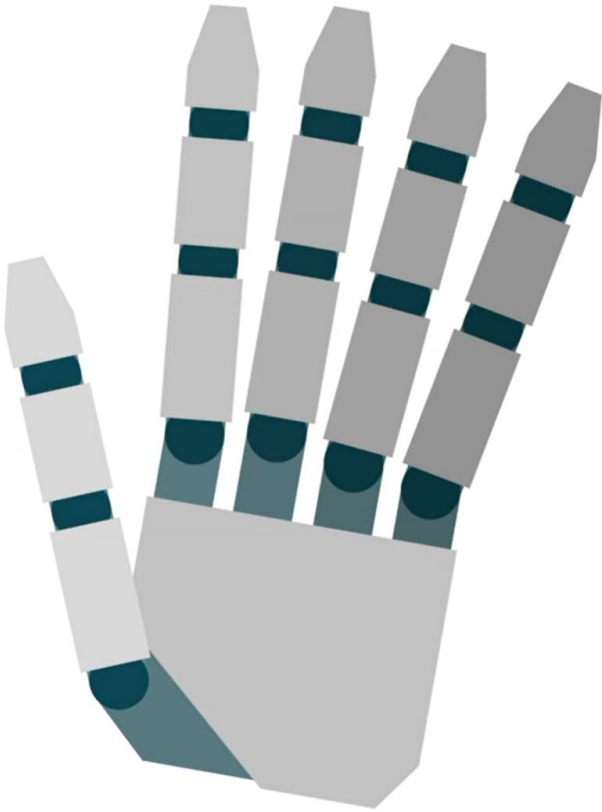
Human Hand and Robotic Hand



Human Hand and Robotic Hand

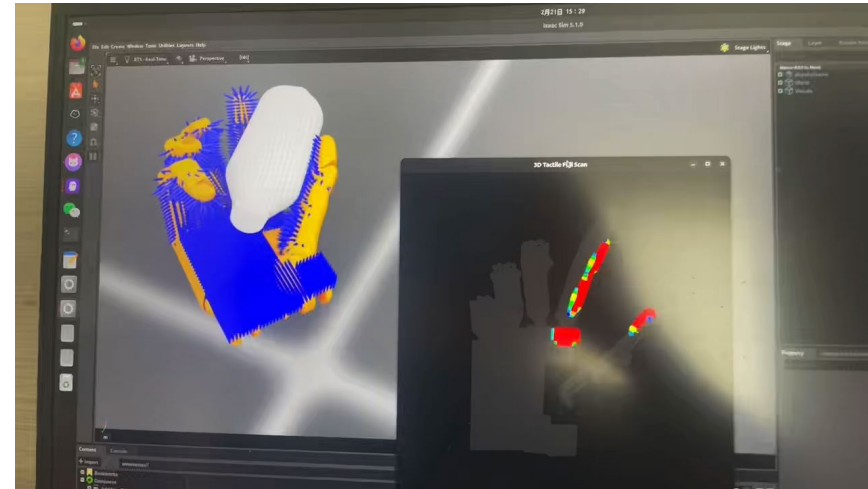


Human Hand and Robotic Hand



Why is challenging to design a dexterous hand ?

- Soft vs Hard
- Tactile design
- Motor
- System integration
- Control algorithm (21DOF)
- Learning new tasks



Why is challenging to design a dexterous hand ?

假设一个机器人有21个自由度，每一个自由度都是0到360度的运动空间，我们现在1度为精度范围，请问这个机器人的搜索空间有多大

- Soft vs Hard
- Tactile design
- Motor
- System integration
- **Control algorithm (21DOF)**
- Learning new tasks

🔗 已思考 (用时 8 秒) >

该机器人有21个自由度，每个自由度在0°到360°范围内以1°为精度，因此每个自由度有360种可能的状态。总搜索空间大小为所有自由度的组合数，即：

$$360^{21}$$

这是一个巨大的数字，用科学记数法表示为约 4.8×10^{53} 。具体计算如下：

$$\log_{10}(360^{21}) = 21 \times \log_{10}(360) \approx 21 \times 2.5563 = 53.6823$$

所以 $360^{21} \approx 10^{0.6823} \times 10^{53} \approx 4.81 \times 10^{53}$ 。

4. 随机猜中的概率

- 双色球头奖概率约为 1.7×10^{-7} 。你的机器人随机一个姿态，正好是目标姿态的概率是 2×10^{-54} 。
- 这个概率比连续中 7 次双色球头奖 ($(1.7 \times 10^{-7})^7 \approx 4.5 \times 10^{-48}$) 还要低 100 万倍！也就是说，你要连续中 8 次双色球头奖，才差不多有这么大的机会。

What is robotic grasping and manipulation?



BBC Documentary: The incredible human hand

What is robotic grasping and manipulation?



What is robotic grasping and manipulation?



“抓取手边的物体并录制视频发给我”

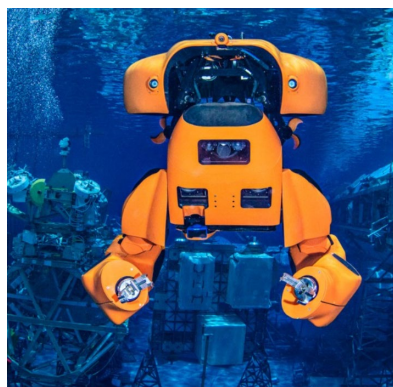
What is robotic grasping and manipulation?



What is robotic grasping and manipulation?



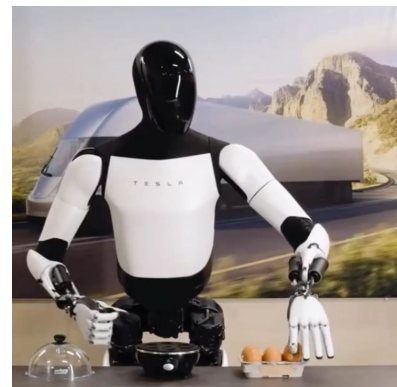
太空探索



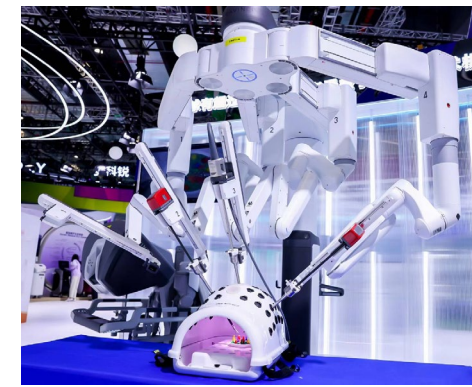
深海勘探



抗震救灾



家庭服务



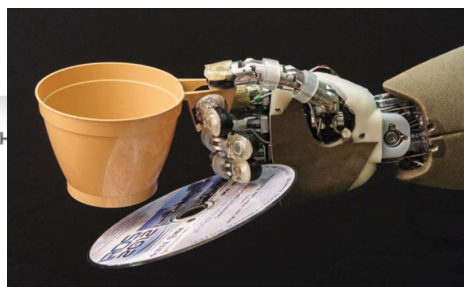
医疗健康

Science

HOME > SCIENCE > VOL. 364, NO. 6446 > TRENDS AND CHALLENGES

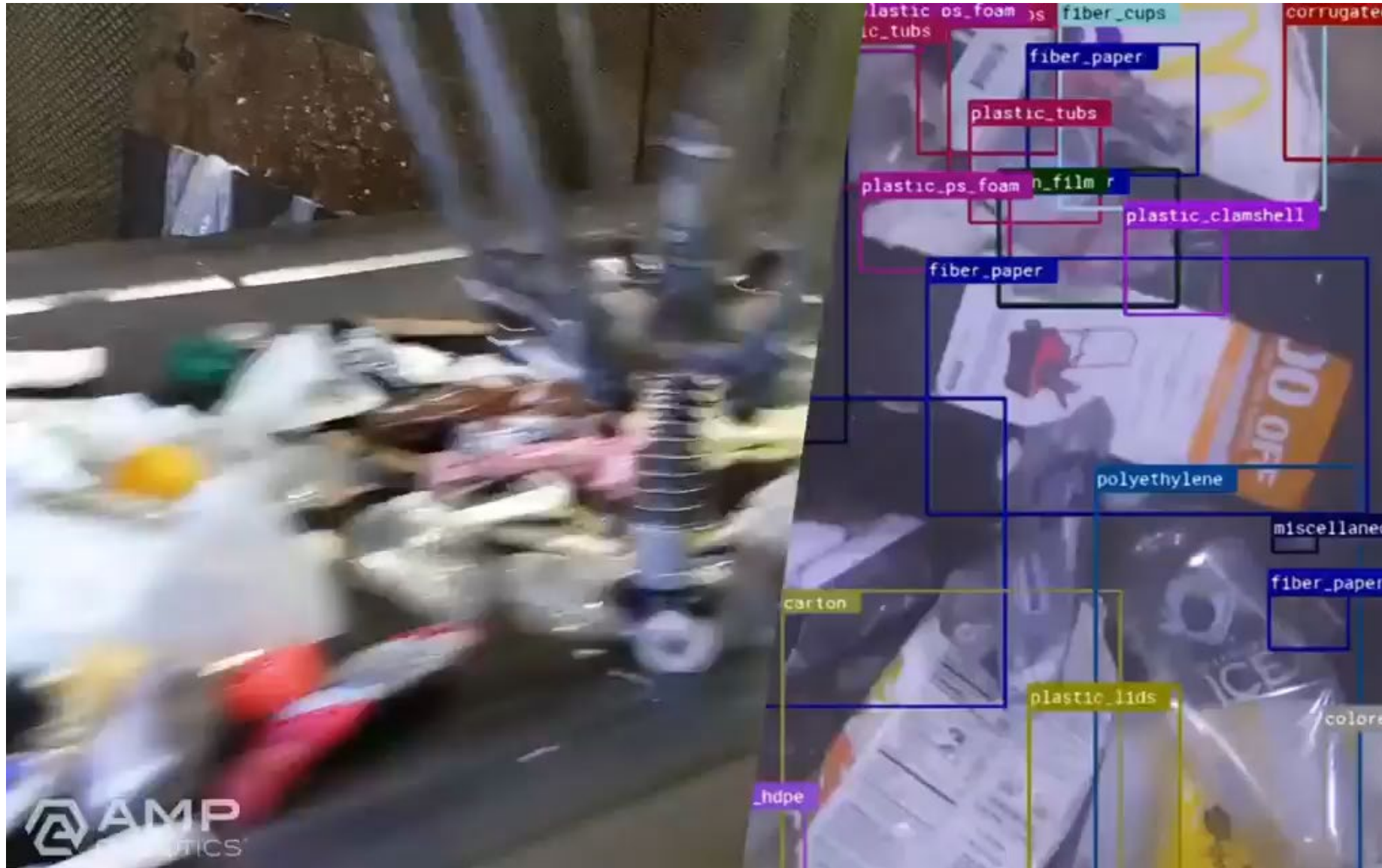
REVIEW

Trends and challenges in robot manipulation



目标：实现机器人灵巧技能作业

What is robotic grasping and manipulation?



What is robotic grasping and manipulation?

Dexterous Robot System for Precise Origami Operations

Hiroto Kitamori¹, Chenyu Dong¹, Masaru Takizawa², Shuya Watanabe¹
Jun Shintake¹, Kohei Kimura¹, and Shunsuke Kudoh¹



¹Graduate School of Informatics and Engineering
at the University of Electro-Communications

²Graduate School of Information Systems
at the University of Electro-Communications

What is robotic grasping and manipulation?

Vision-Based In-Hand Manipulation of Variously Shaped Objects via Contact Point Prediction

Yuzuka Isobe¹, Sunhwi Kang², Takeshi Shimamoto², Yoshinari Matsuyama²,
Sarathak Pathak¹ and Kazunori Umeda¹

1. Chuo University, Japan

2. Panasonic Connect Co., Ltd., Japan

What is robotic grasping and manipulation?

SDP-RRT: A New Method Combining Single-point Push and Double-point Complete Push for Partially Observable Scenarios

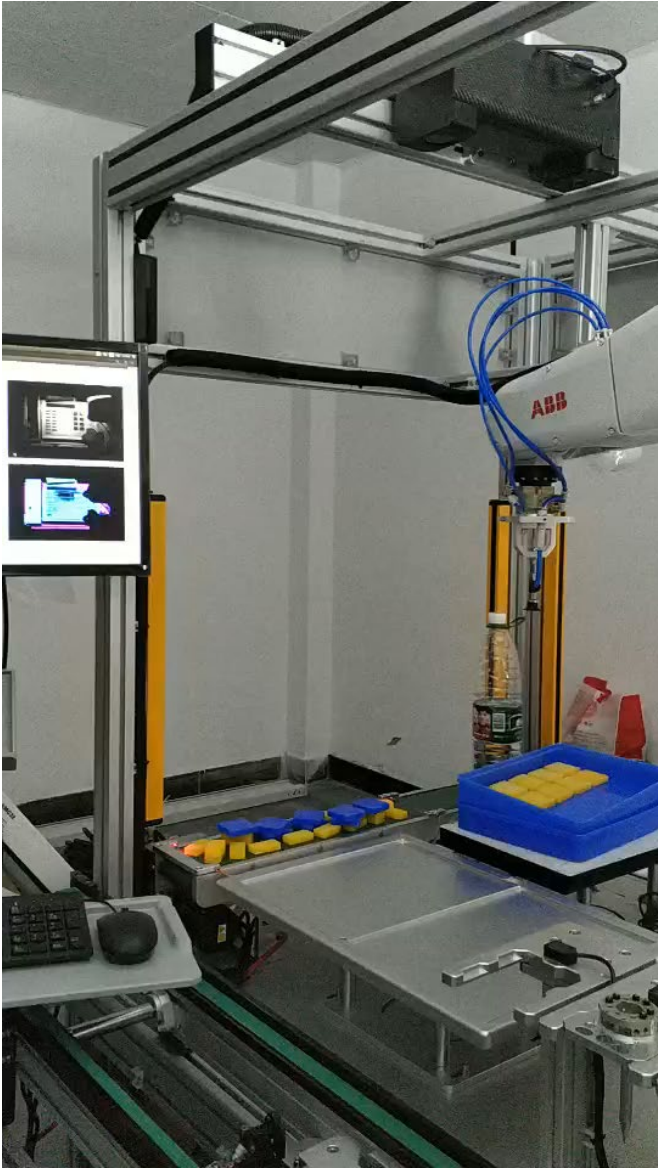
Jiawei Zhang, Chengchao Bai, and Jifeng Guo
Harbin Institute of Technology



What is robotic grasping and manipulation?



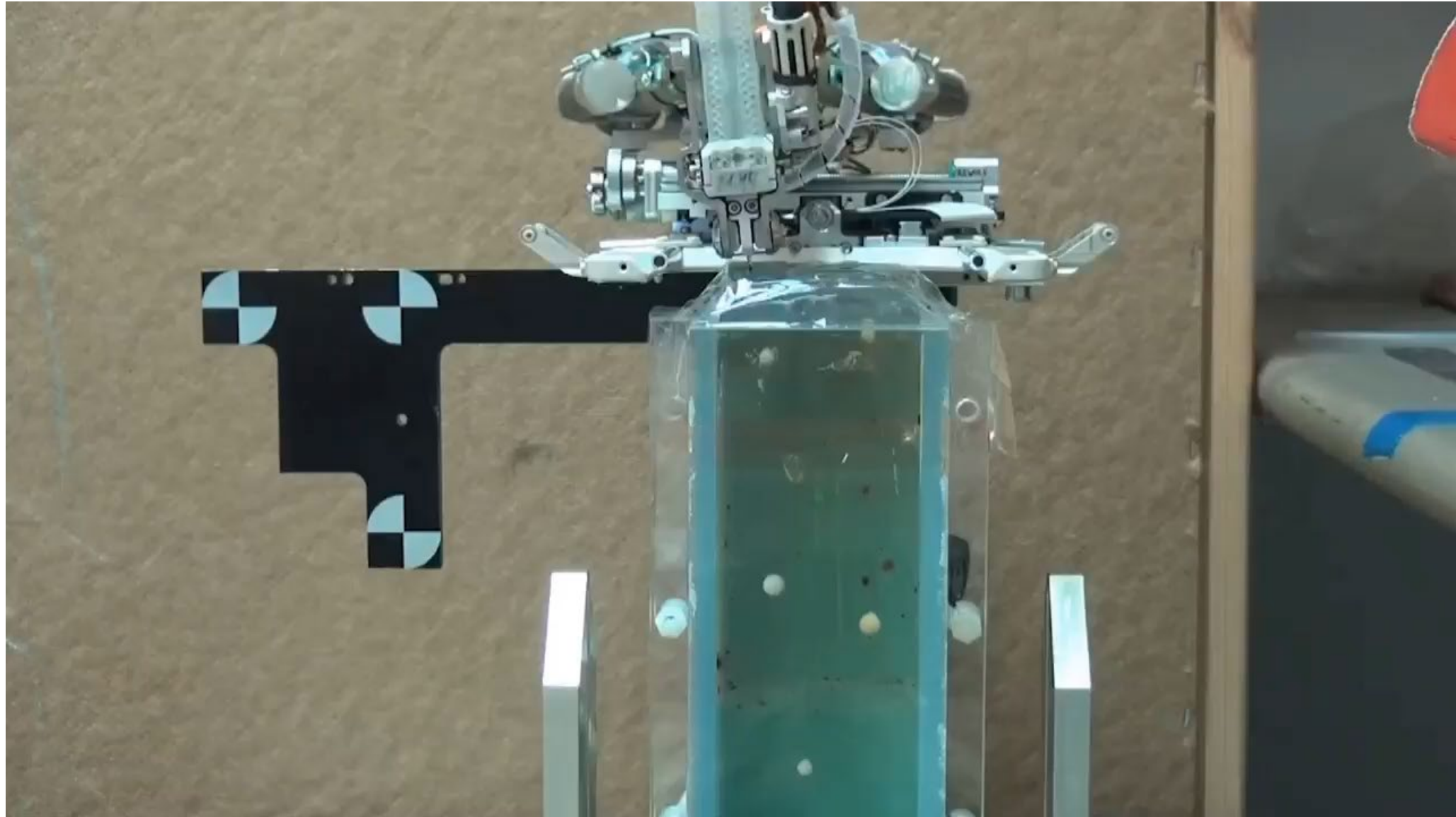
What is robotic grasping and manipulation?



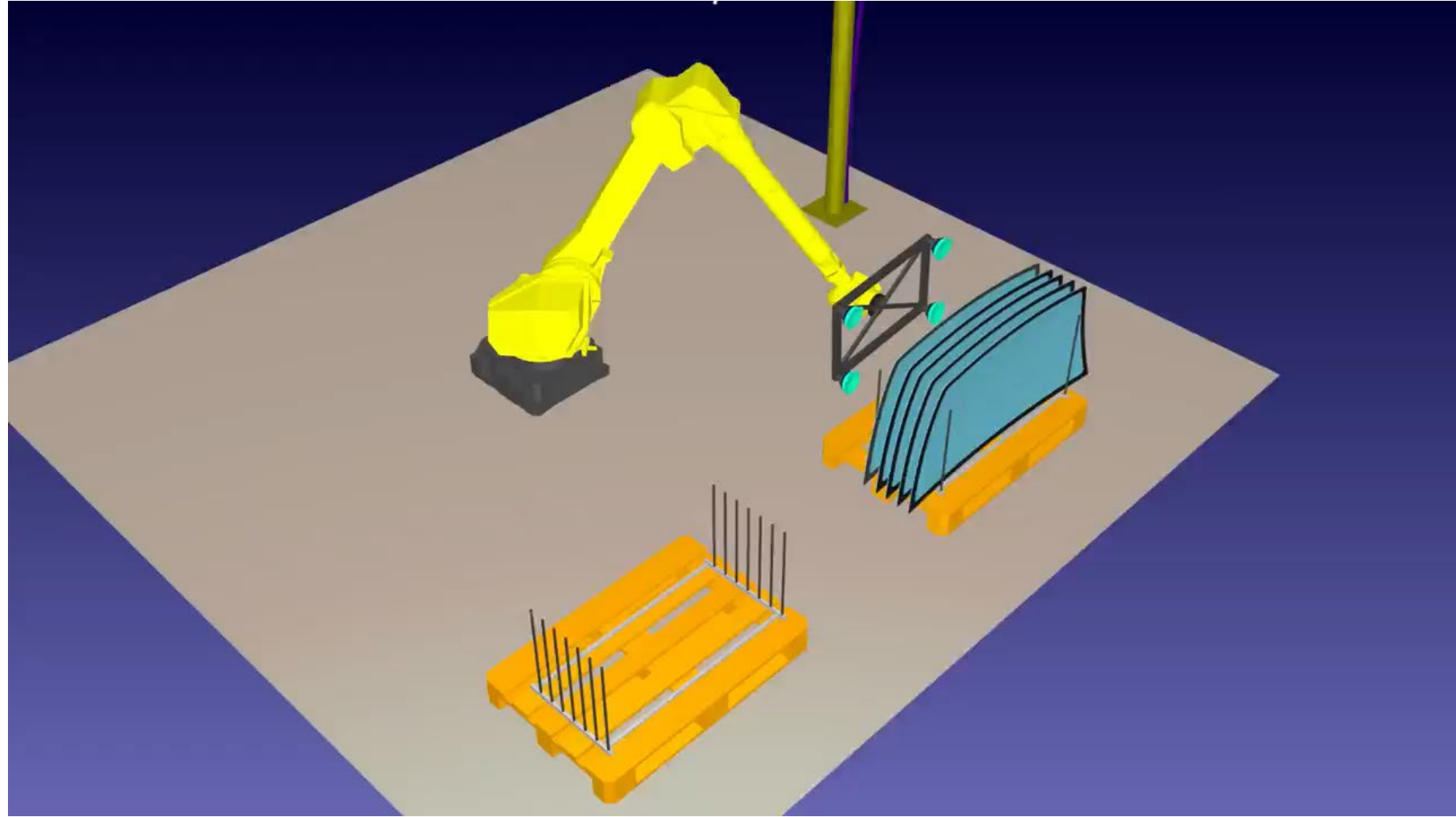
What is robotic grasping and manipulation?



What is robotic grasping and manipulation?



What is robotic grasping and manipulation?

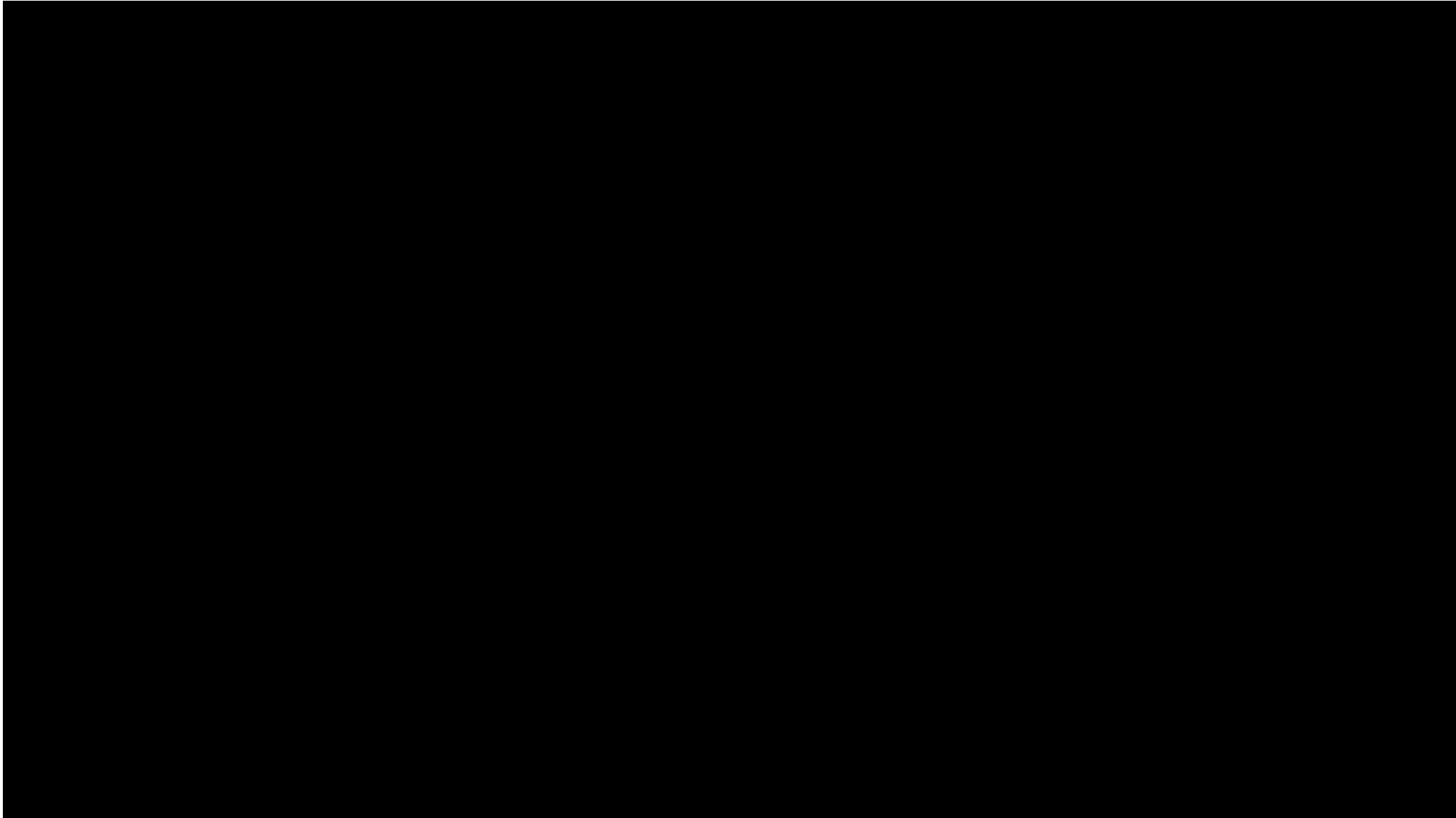


What is robotic grasping and manipulation?

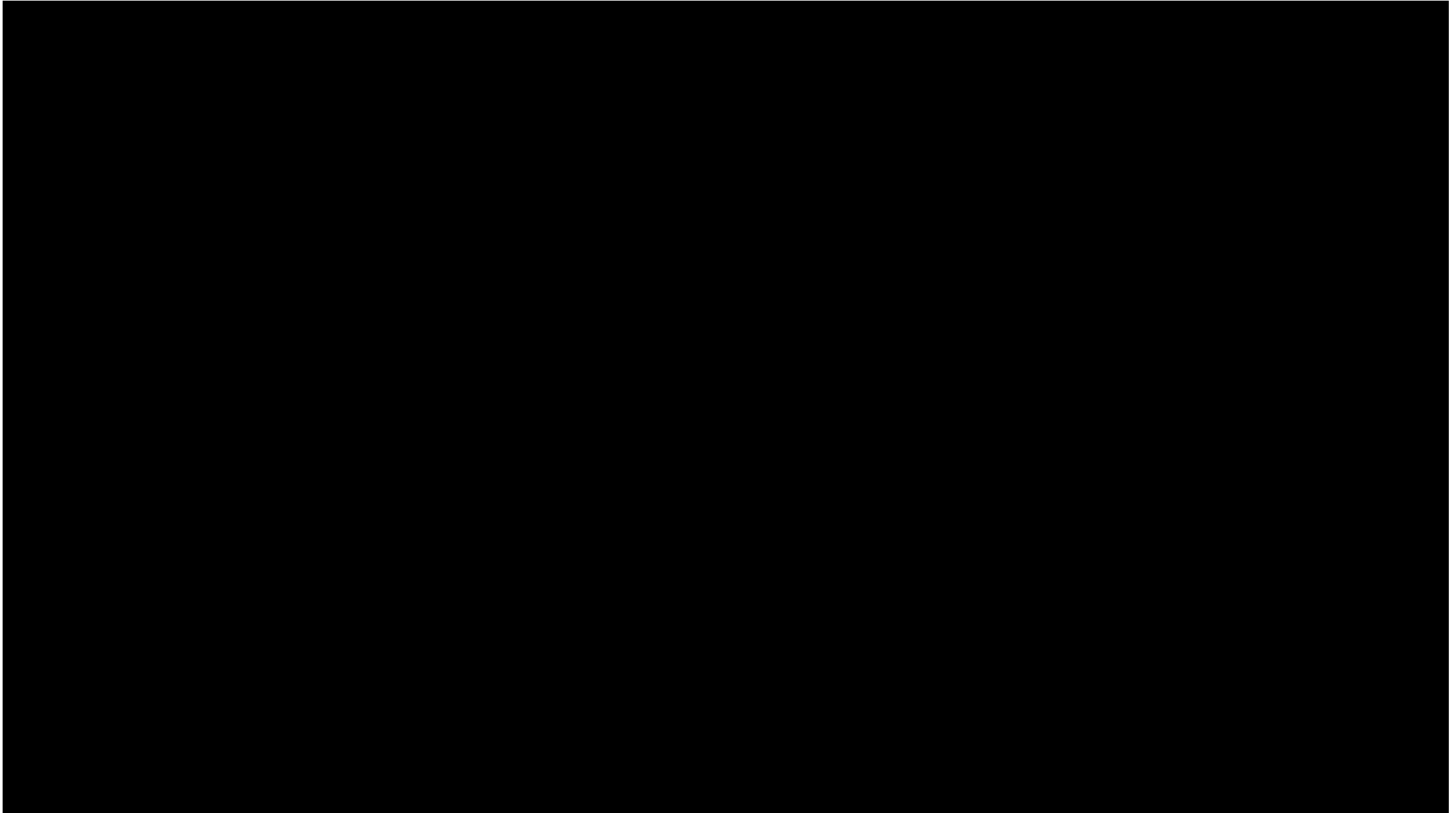


**Kawasaki Robotics ZX300S Robots
Barrel Handling at
Speyside Cooperage Limited, Scotland**

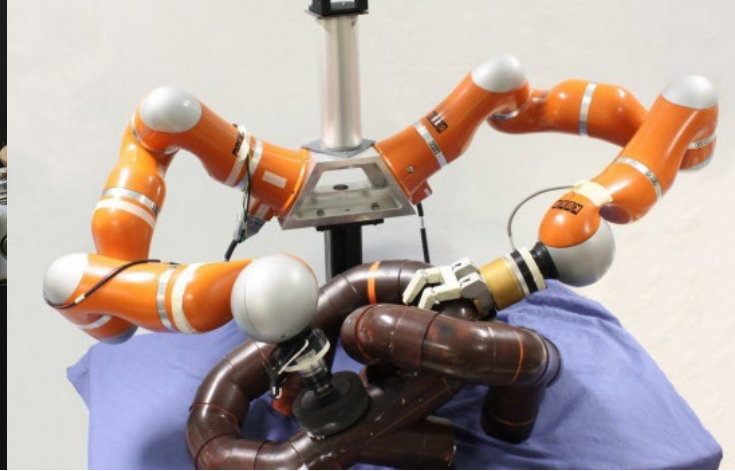
What is robotic grasping and manipulation?



What is robotic grasping and manipulation?



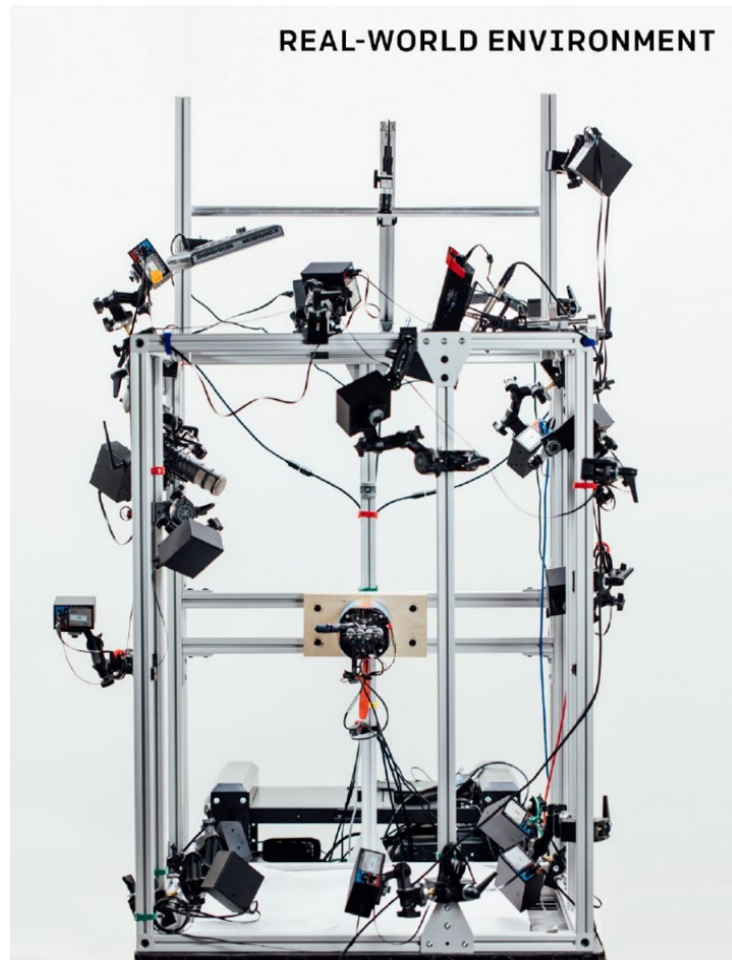
What is robotic grasping and manipulation?



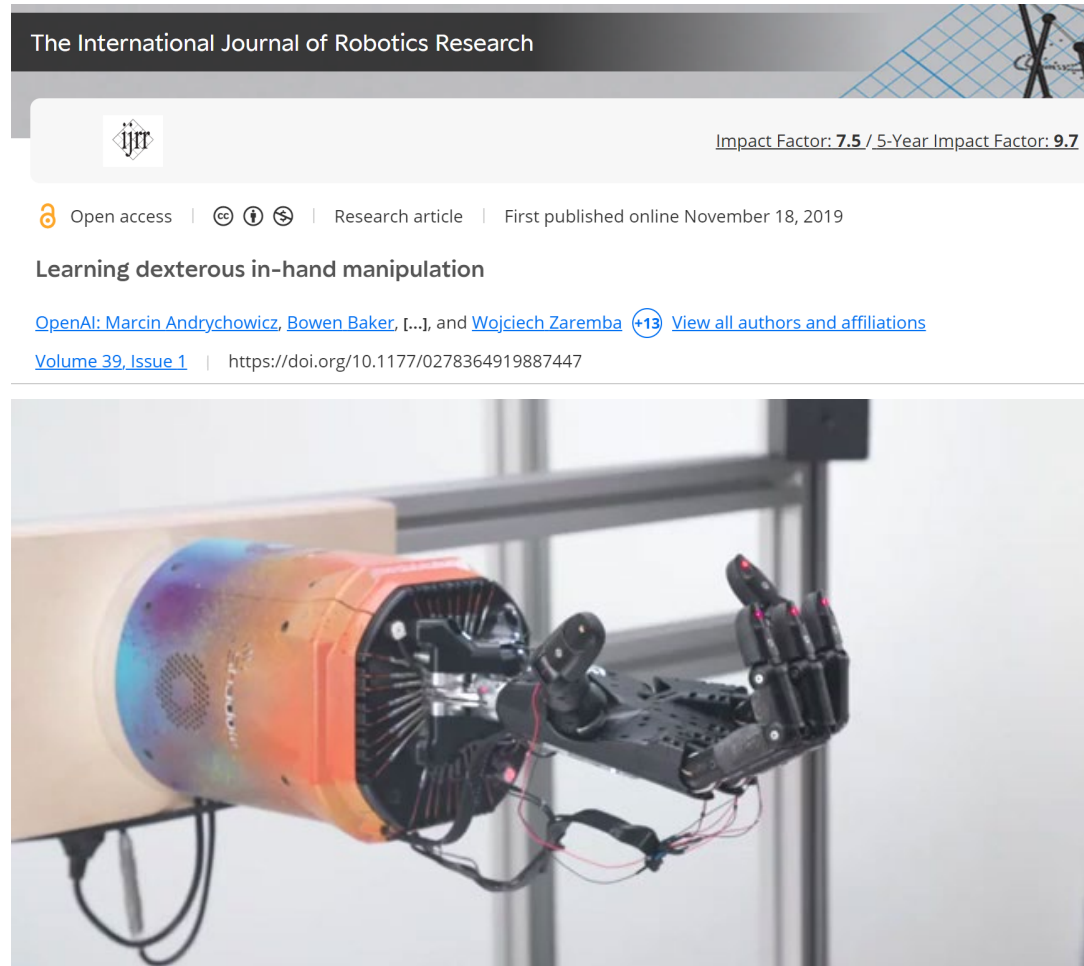
What is robotic grasping and manipulation?



What is robotic grasping and manipulation?

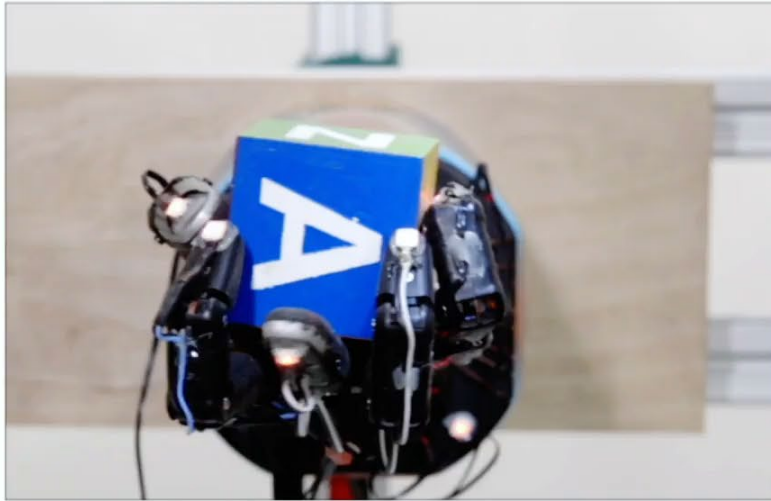


16 PhaseSpace tracking cameras,
and 3 Basler RGB cameras

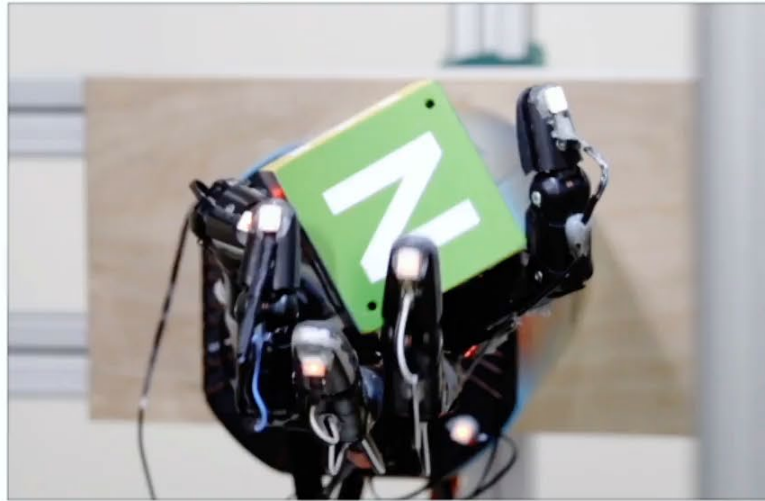


OpenAI, Dexterous Manipulation

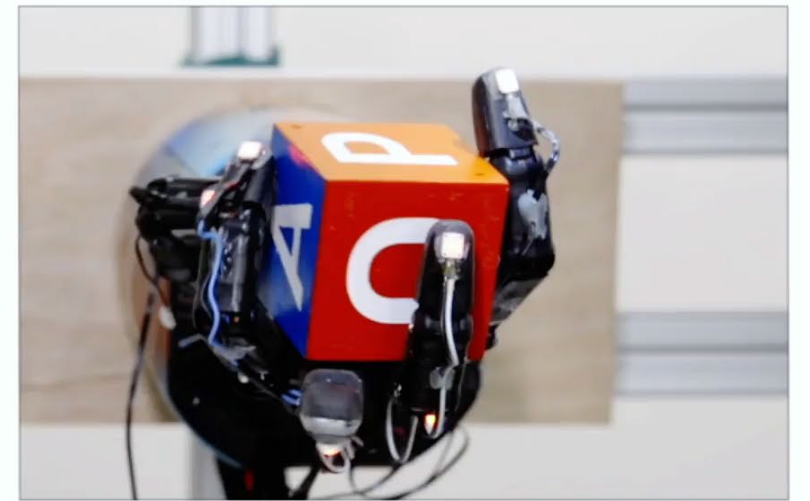
What is robotic grasping and manipulation?



FINGER PIVOTING



SLIDING



FINGER GAITING

What is robotic grasping and manipulation?



What is robotic grasping and manipulation?

EPFL

LASA
Learning Algorithms and
Systems Laboratory

Exploiting Kinematic Redundancy for Robotic Grasping of Multiple Objects

Kunpeng YAO and Aude BILLARD

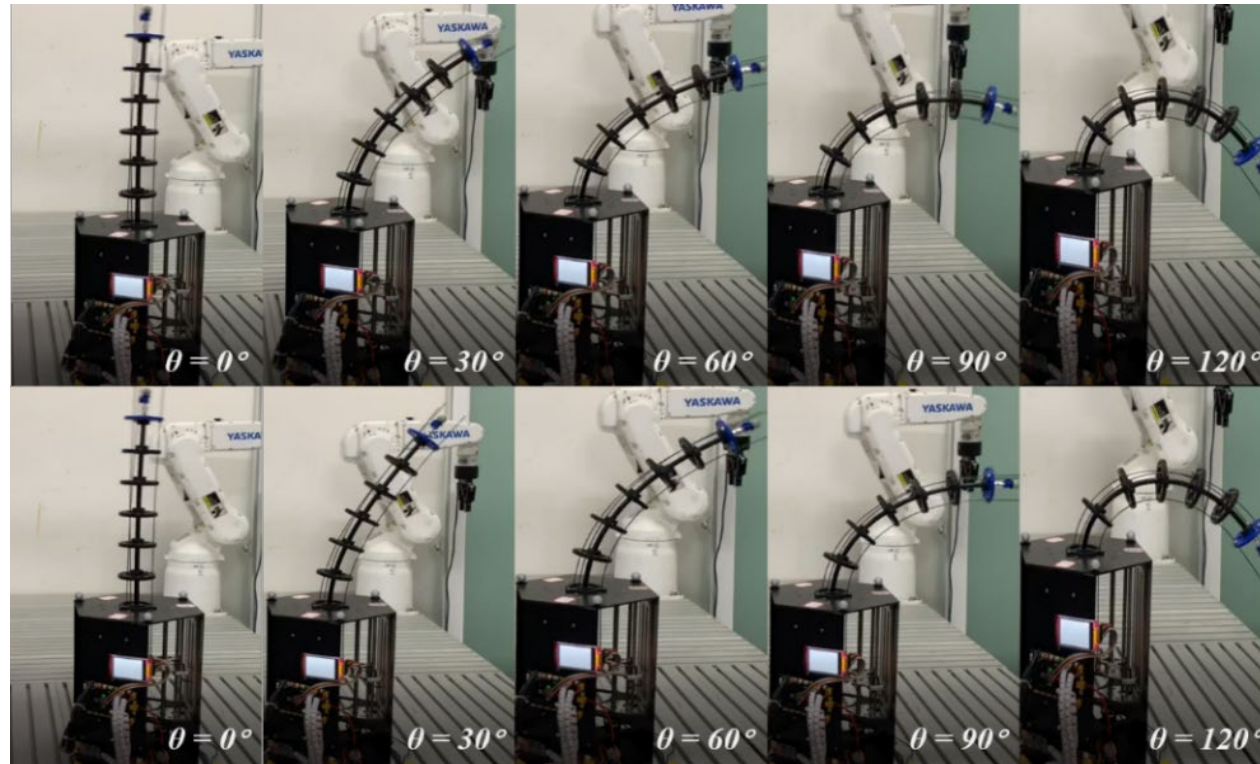
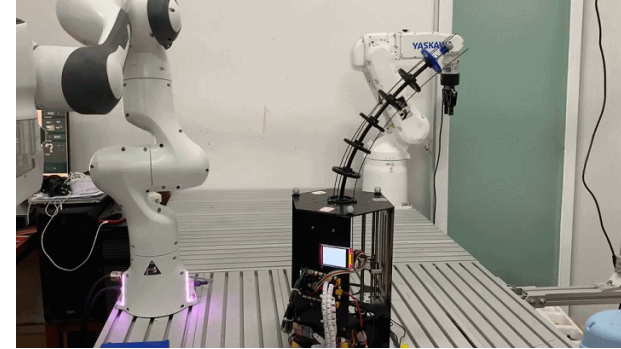
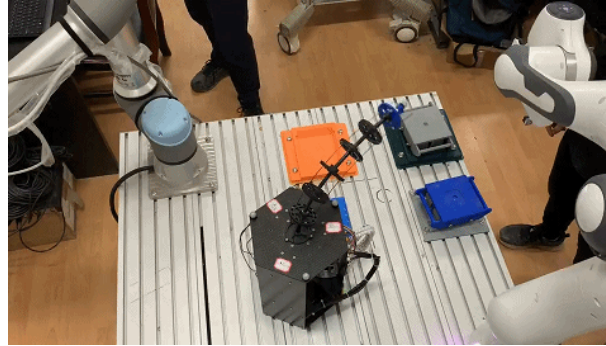
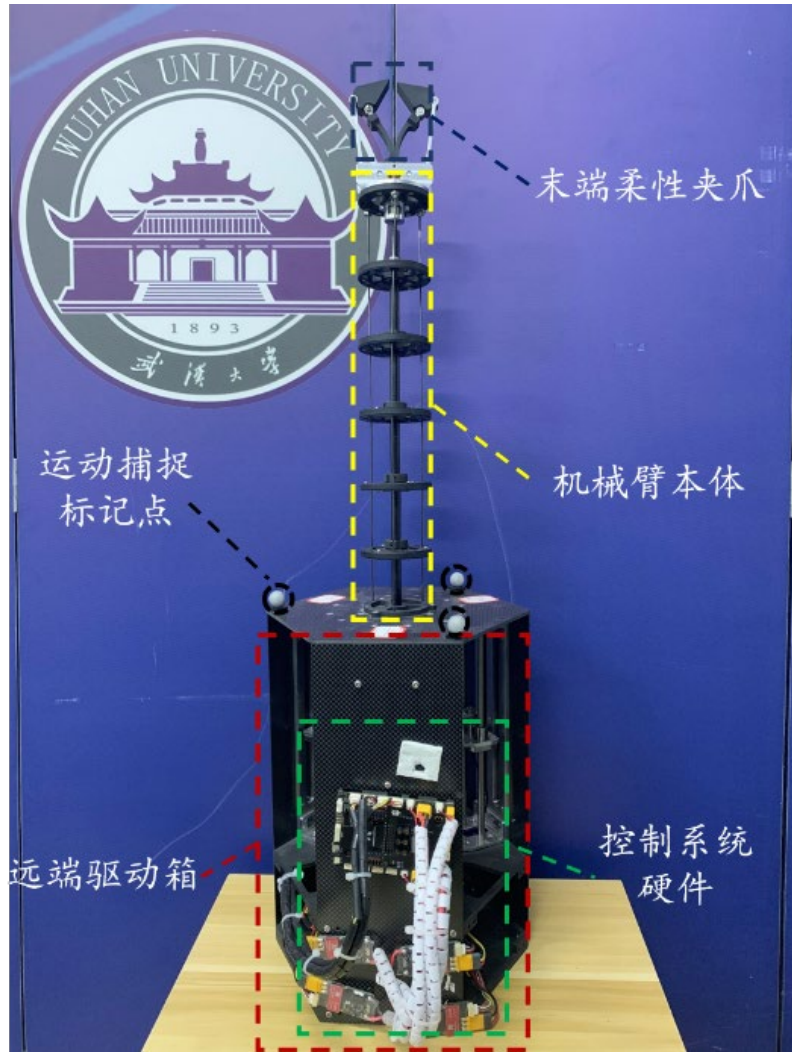
Learning Algorithms and Systems Laboratory (LASA)

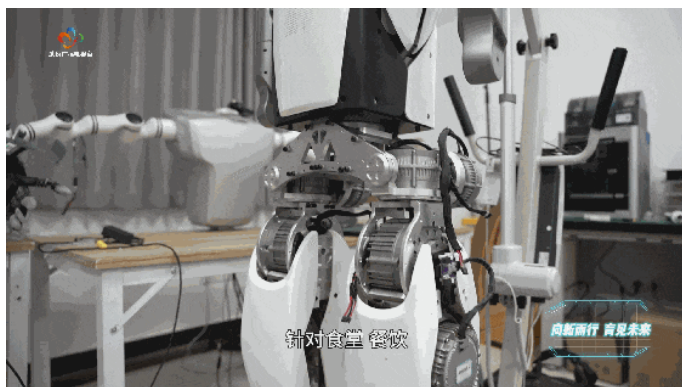
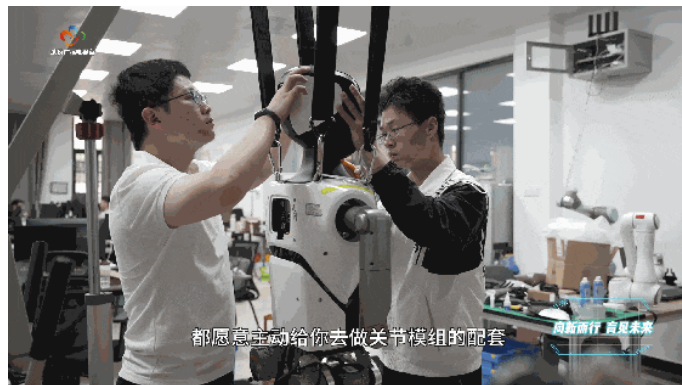
École Polytechnique Fédérale de Lausanne (EPFL)



This project is supported through an ERC Advanced Grant, project ID: 741945, funded by the European Commission.

What is robotic grasping and manipulation?

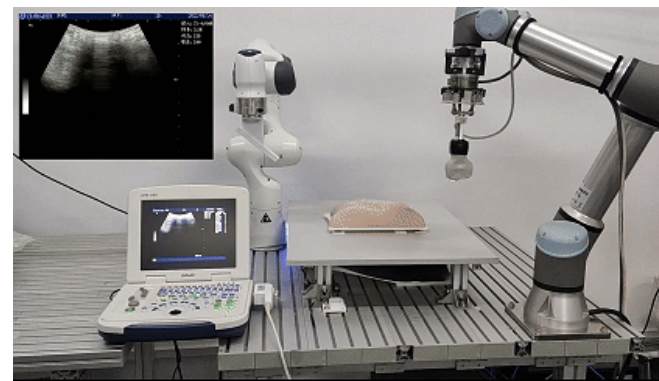




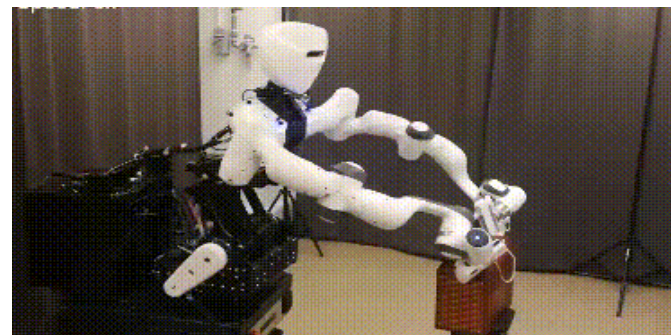
What is robotic grasping and manipulation?



机器人炒菜



双臂协作超声穿刺

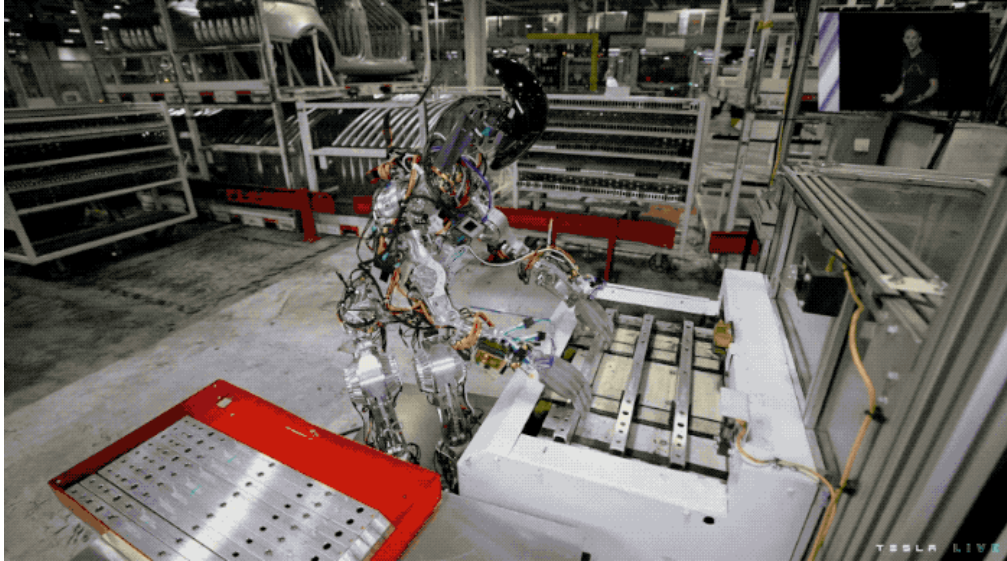


双臂协作人机交互

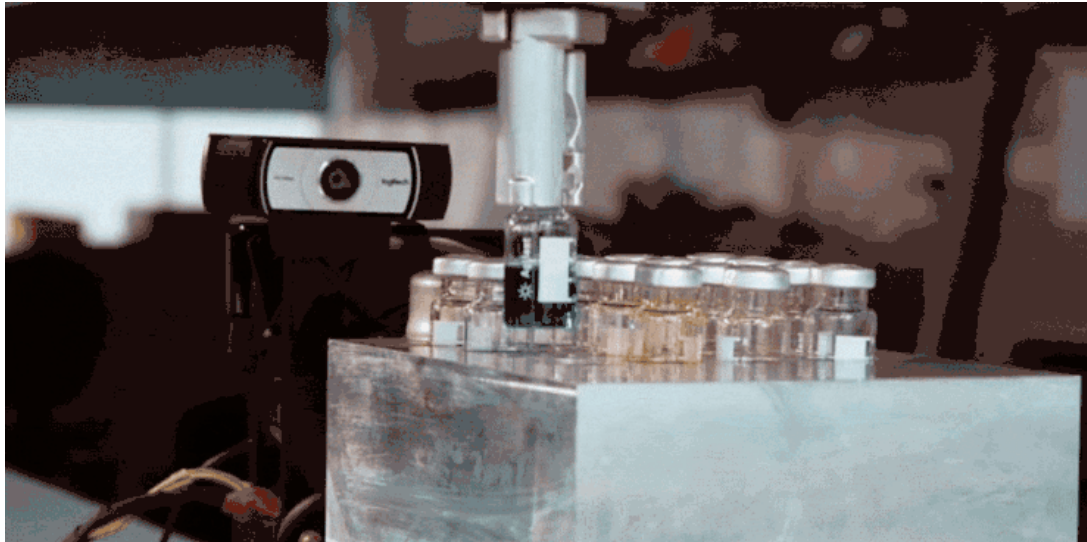


机器人协作做咖啡

What is robotic grasping and manipulation?



What is robotic grasping and manipulation?



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What is robotic grasping and manipulation?

AS01CH19_Mason ARI 19 February 2018 14:24



*Annual Review of Control, Robotics, and
Autonomous Systems*

Toward Robotic Manipulation

Matthew T. Mason

Robotics Institute and Computer Science Department, Carnegie Mellon University, Pittsburgh,
Pennsylvania 15213, USA; email: matt.mason@cs.cmu.edu

What is robotic grasping and manipulation?

Definition 1 (etymological). Manipulation refers to the activities performed by hands.

The etymological definition gives little guidance in determining whether an activity is manipulation or not, although it captures a shared understanding and point of departure for the discussion. It helps us ask the right question: What are the intrinsic defining criteria of manipulative activities?

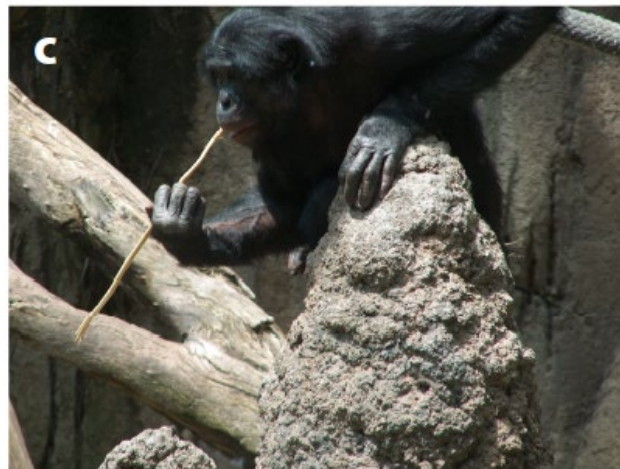
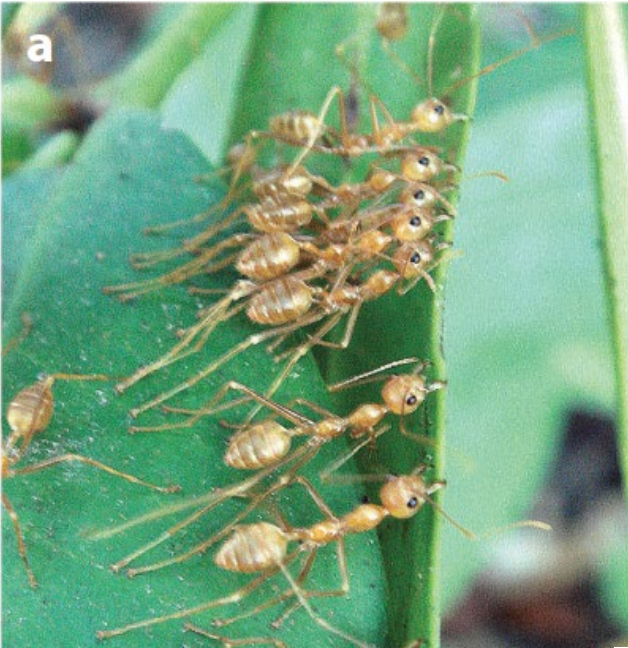
Definition 2 (genus/differentia, ends only). Manipulation is when an agent moves things other than itself.

Definition 3 (genus/differentia, ends and means). Manipulation is when an agent moves things other than itself through selective contact.

Definition 4 (bottom up). Manipulation is pick-and-place manipulation plus in-hand manipulation plus mechanical assembly plus. . . .

Definition 5. Manipulation refers to an agent's control of its environment through selective contact.

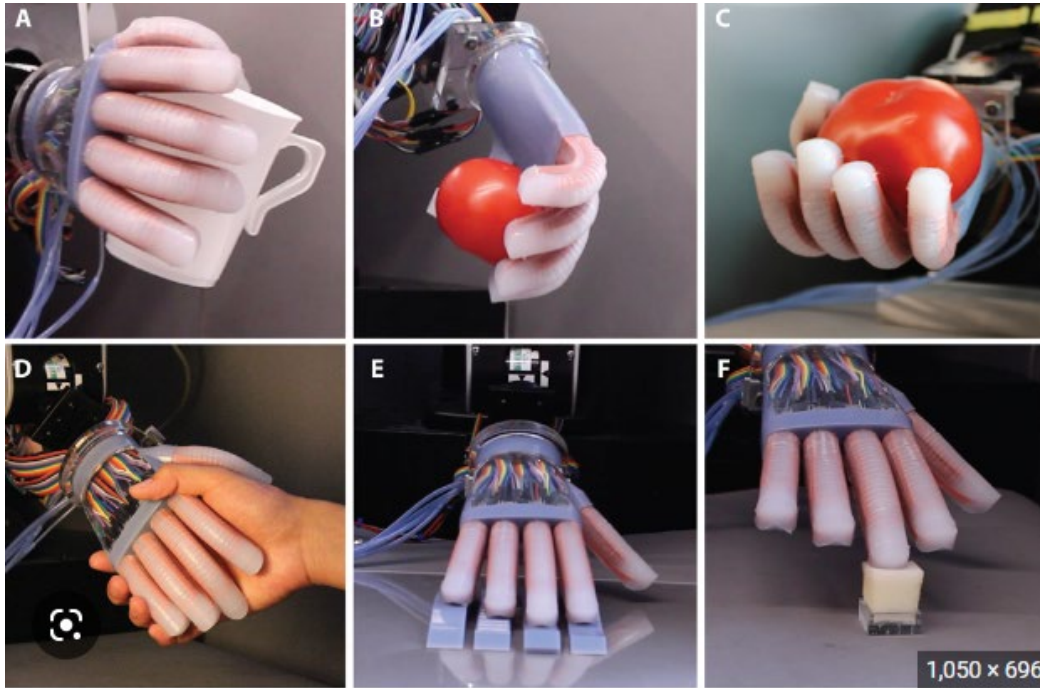
Example of animal manipulation



**More interesting
examples?**

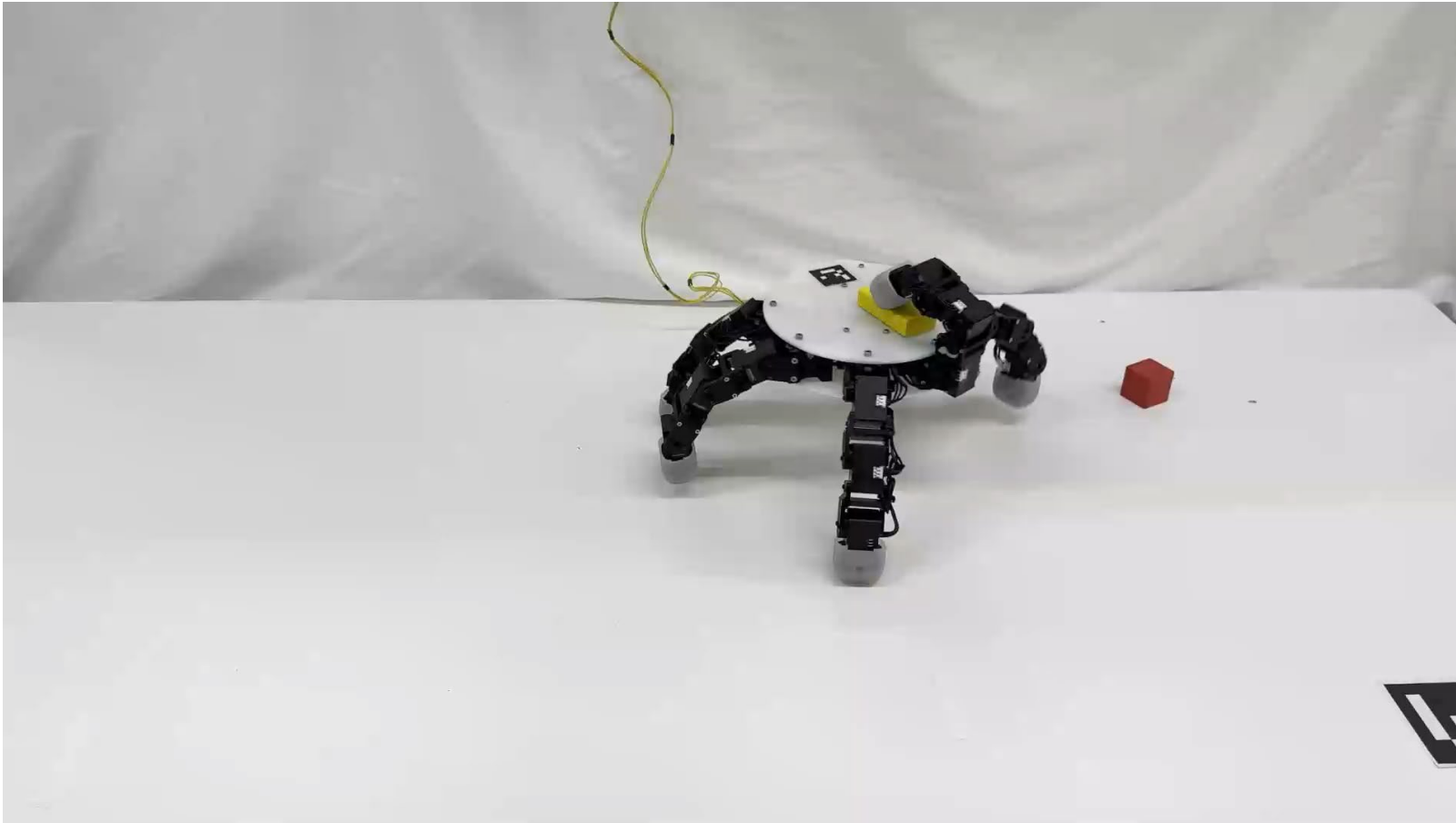
The project of this year

- **Design a soft hand that can grasp interesting stuff**



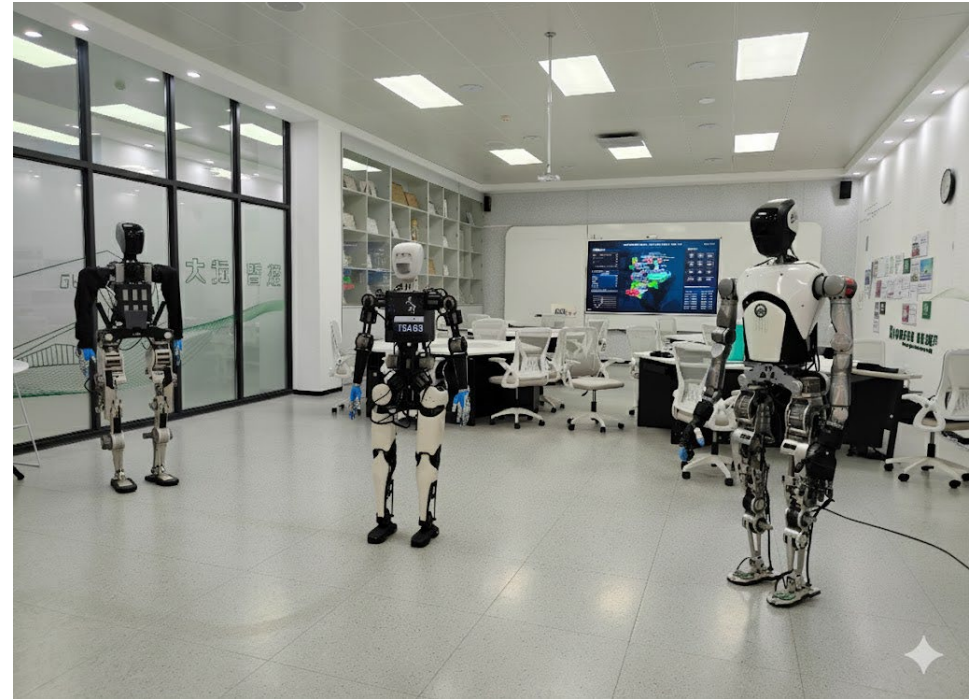
The project of this year

- Design a soft hand that can crawling



Guideline for the project

- Think outside the box
- Report (30%), Presentation(30%) and Demo (40%)
- Use AI tools before you ask for help
- If you want to continue the research after the course, please contact us ASAP



武汉大学3D打印人形机器人五指灵巧手，问鼎2025德国红点奖

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IOP Publishing

Smart Mater. Struct. 34 (2025) 125030 (13pp)

Smart Materials and Structures

<https://doi.org/10.1088/1361-665X/ae270f>

A compliant, underactuated soft hand inspired by mechanical metamaterials

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Abstract

Grasping and manipulating objects in unstructured environments remain challenging due to uncertainties in object geometry, size, and physical properties. Conventional robotic hands, although capable, often incorporate complex mechanical structures and advanced sensing-control systems. This paper introduces a novel underactuated, tendon-driven soft hand inspired by mechanical metamaterials, designed for adaptive grasping of diverse objects. Each finger features a 3D-printed thermoplastic polyurethane structure integrated with negative Poisson's ratio metamaterial microstructures and passive flexure joints, enabling compliant multi-axis deformation. Experimental validation using 12 objects of varied shapes, sizes, and textures demonstrated robust performance, achieving grasp success rates of 96.3% for enveloping grasps and 88.1% for precision grasps.

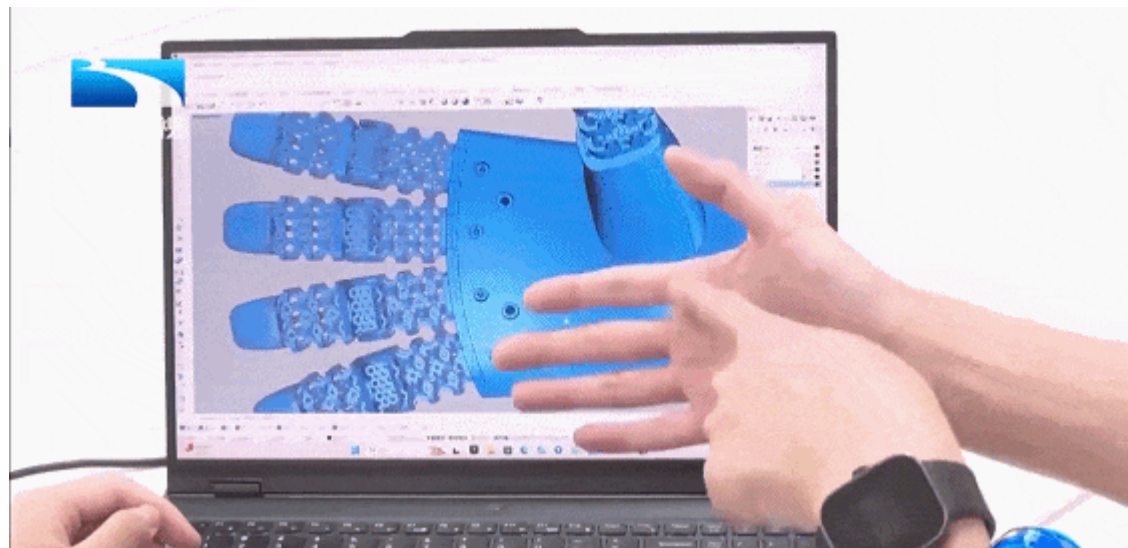
Supplementary material for this article is available [online](#)

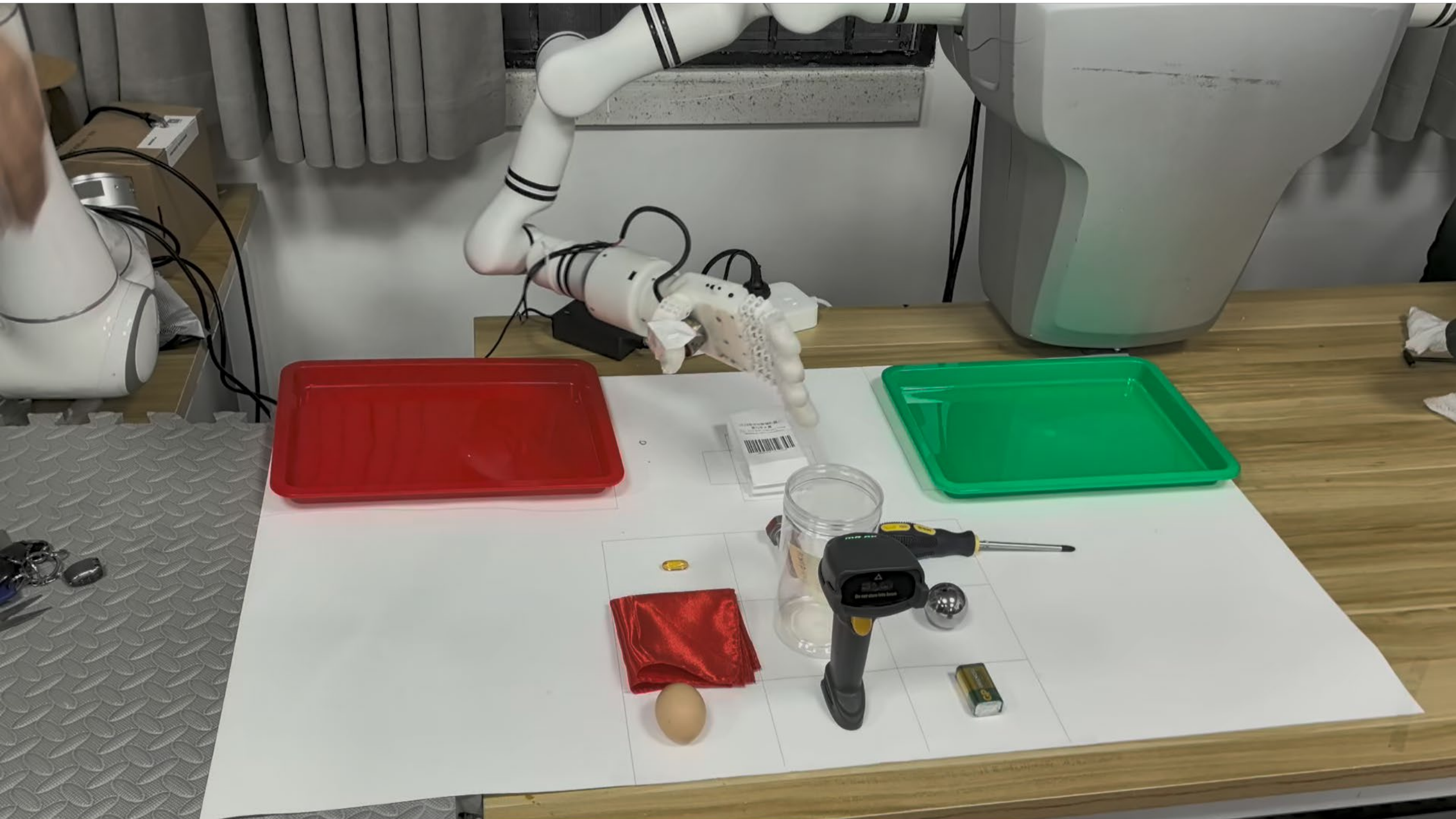
Keywords: underactuated soft hand, mechanical metamaterials, negative Poisson's ratio, tendon-driven actuation, adaptive grasping

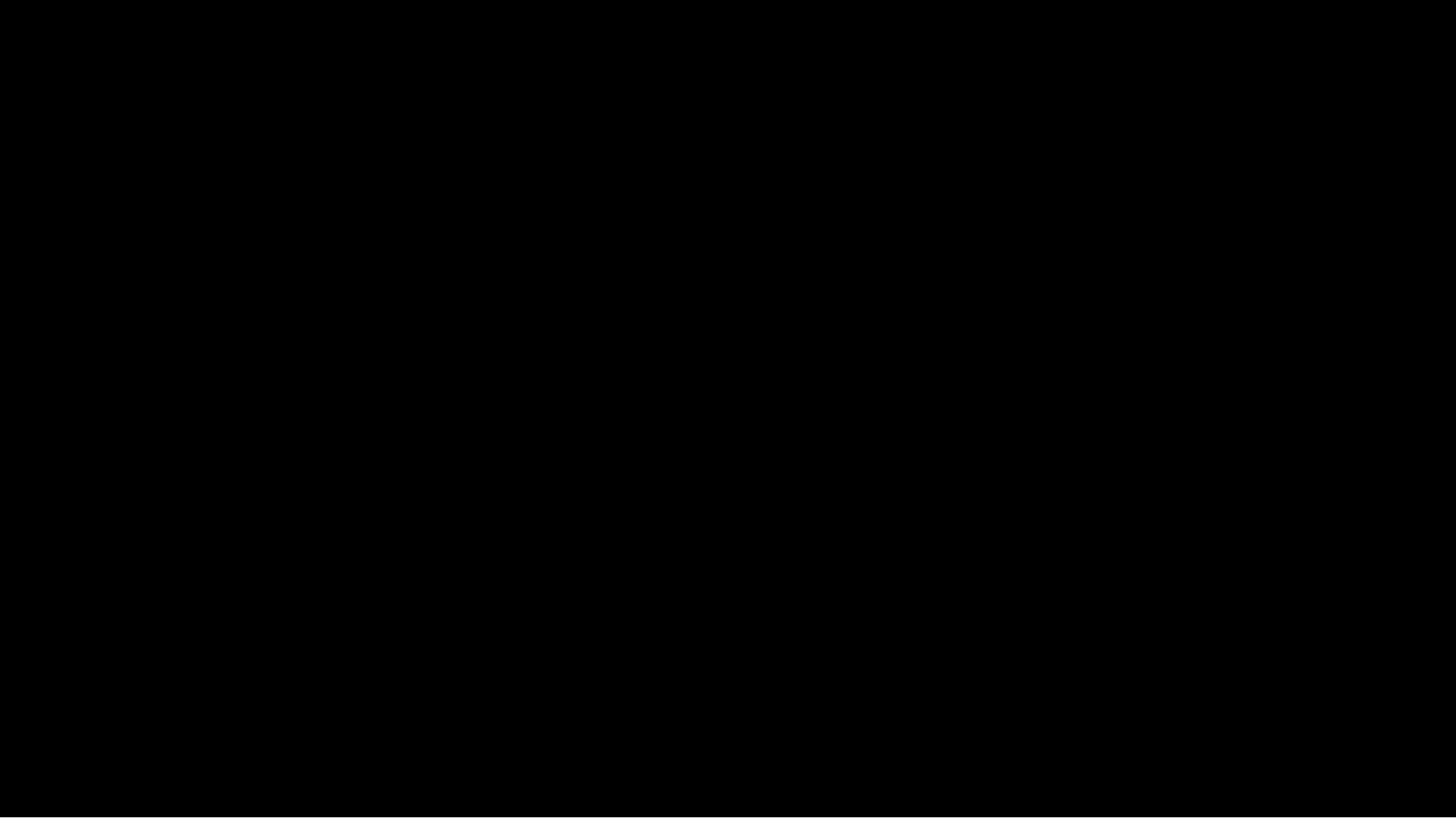


A











武汉大学首台

人形机器人—“天问” 半身人形轮式机器人—“远游”

Wuhan University's first humanoid robot - "TIANWEN"
Semi-humanoid wheeled robot - "YUANYOU"



天问
WHU humanoid NO.1
头部:3自由度
视觉传感器
视觉摄像头
手臂:6自由度*2
手部:5自由度*2
全身:6自由度*2
腰部:4自由度
左右±45° 前后±90°
关节最大速度:720°/s
关节最大扭矩:40Nm
腰部关节:10°-150°
腿部:3自由度*2
Head: 3 degrees of freedom (DOF)
Effector sensor
Visual camera
Arm: 6 degrees of freedom (DOF) * 2
Hand: 5 degrees of freedom (DOF) * 2
Full body tactile sensor
Waist: 4 degrees of freedom (DOF)
Left and right ±45° front and back ±90°
Maximum joint speed of 720°/s
Maximum joint torque of 40 Nm
Waist joint: 10° to 150°
Leg: 3 degrees of freedom (DOF) * 2

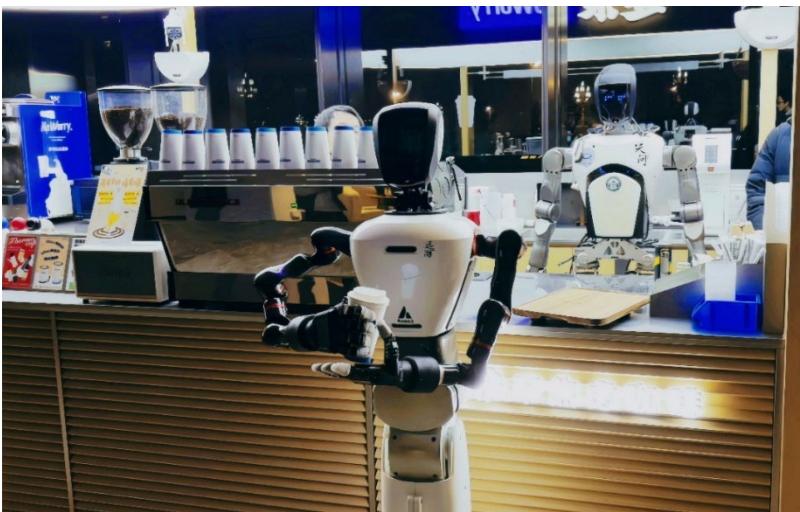






The First Meeting of Hubei in the New Spring





长江日报

中共武汉市委
主 编：王 勇
2025年2月26日 星期三
农历丙午年正月廿二

影响有影响的人

总编：张 晓 明
副总编：张 晓 明
副总编：张 晓 明

中共中央国务院印发 《国家突发事件总体应急预案》

新华社北京2月25日电 新华社北京2月25日电 中共中央、国务院日前印发《国家突发事件总体应急预案》。《预案》是党中央、国务院为应对突发事件的总体预案，是应急管理工作的总纲。《预案》的印发，是党中央、国务院在总结近年来应对突发事件的经验教训，深刻汲取教训，全面加强应急管理工作的基础上，对突发事件应对工作作出的重大决策部署。《预案》的印发，是党中央、国务院在总结近年来应对突发事件的经验教训，深刻汲取教训，全面加强应急管理工作的基础上，对突发事件应对工作作出的重大决策部署。《预案》的印发，是党中央、国务院在总结近年来应对突发事件的经验教训，深刻汲取教训，全面加强应急管理工作的基础上，对突发事件应对工作作出的重大决策部署。

公开亮相20天后就试岗 人形机器人“天问”上咖啡店实习

郭元强赴东西湖区走访调研企业
加大帮扶力度 增强企业信心
奋力实现一季度良好开局

【记者报道】2月25日，在东西湖区一家名为“NO WORRY COFFEE”的咖啡店，一款名为“天问”的人形机器人正在接受培训。这款由武汉大学研发的机器人，身高1.75米，体重120公斤，外观酷似真人。它正在学习咖啡制作流程，包括磨豆、萃取、打奶泡等。在培训过程中，它表现得非常认真，动作流畅。培训结束后，它将在该咖啡店进行为期20天的试岗。这款机器人的试岗，是武汉市推动人工智能与实体经济深度融合的又一重要举措。郭元强表示，武汉市将加大对企业的帮扶力度，增强企业信心，奋力实现一季度良好开局。

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你喝的咖啡 可能是机器人做的 武汉大学人形机器人 到咖啡店“实习”了

2月25日
武汉大学研发的基于咖啡场景的
人形机器人“天问”
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